

The challenge wasn't just the value of these instruments, but also how the information was organized—or more accurately, disorganized. Despite their sophisticated financial models, Lehman lacked a clear, unified structure for their data. By Monday morning, the bank declared bankruptcy in what became the largest in U.S. history.

While many factors contributed to Lehman's collapse, one lesson rang clear: in finance, how you organize your data is just as important as the data itself.

This idea sits at the heart of algorithmic trading. Before a single line of trading logic can be written, you need effective ways to organize financial information: stock prices, trading signals, portfolio holdings, market indicators, and more. This is precisely what Python's data structures provide.

Data structures are simply organized containers for information. Think of them as different types of filing systems, each with unique strengths for particular tasks. Just as a library organizes books by subject and author to make finding information easier, Python's data structures organize data in ways that make it more accessible and easier to manipulate.

In this chapter, we'll explore four fundamental data structures in Python:

1.

Lists: Ordered collections of items, perfect for time series data like price histories

2.

Tuples: Immutable (unchangeable) sequences, ideal for data that shouldn't be modified

3.

Dictionaries: Key-value pairs that allow you to look up information by name

4.

Sets: Collections of unique items, useful for filtering and finding commonalities

By the end of this chapter, you'll understand how to use these structures to organize financial data effectively—a crucial foundation for building successful trading algorithms.